

classical critical-path cost

condition that eliminates it

order log
on every message

network

O: delivery in one global order
→ replay without a log

coordinated snapshot
(markers + channel state)

O + R: consistent, uncoordinated
cut at timestamp T , channels empty

output held for extra
durability round trips

R + D: durability completes inside
the barrier → no added round trip

local non-determinism
(time · randomness · threads)

host

L: user-space determinism shim
→ byte-identical replay